

F-Harmonic Forms on Symplectic Lie Groups: Mathematical Background

Introduction

Let (M, ω) be a 6-dimensional symplectic Lie group, let $\mathfrak{g}$ its Lie algebra. We compute its Lie algebra cohomology and symplectic cohomology analogues. In general, for a 6-dimensional symplectic manifold, from any 3-form ϕ on M , or better any $\phi \in \bigwedge^3 \mathfrak{g}^*$, one can construct an endomorphism $K(\phi)$ of TM , another 3-form $F(\phi)$ and a scalar function $Q(\phi)$. The endomorphisms F, K, Q are homogeneous polynomials in ϕ of degree 2, 3 and 4, respectively.

We say ϕ is F harmonic if $d\phi = 0$ and $dF(\phi) = 0$. We want to characterize the set of F harmonic 3-forms. Also, we want to consider the following evolution equation of 3-forms:

$$\frac{\partial_t \phi}{\omega} = d\Lambda_\omega dF(\phi),$$

where Λ_ω is the Lefschetz operator of contraction with respect to the symplectic form ω . We want to verify the short-time existence and uniqueness.

Background

Lie Algebras and Structure Constants

Let G be a Lie group of real dimension n . All the left-invariant data on G are determined by their value at the identity, hence they can be viewed as an element of the Lie algebra $\mathfrak{g}$ or other algebraic constructions built from $\mathfrak{g}$. After fixing a basis $\{e_1, e_2, \dots, e_n\}$ of $\mathfrak{g}$, the isomorphism class of $\mathfrak{g}$ is determined by the structure constants $\{c_{ij}^k\}_{1 \leq i, j, k \leq n}$, which are defined by:

$$[e_i, e_j] = c_{ij}^k e_k$$

In many cases, it is more convenient to work with the dual Lie algebra $\mathfrak{g}^*$ and its exterior product:

$$\bigwedge^* \mathfrak{g}^* = \bigoplus_{i=0}^n \bigwedge^i \mathfrak{g}^*$$

Exterior Derivative on Lie Algebras

The structure of $\mathfrak{g}$ can be recovered from the exterior derivative d defined on $\mathfrak{g}^*$, given by:

$$de^k = -\frac{1}{2}c_{ij}^k e^i \wedge e^j$$

where $\{e^1, \dots, e^n\}$ and c_{ij}^k are the dual basis and the structure constants associated to the basis $\{e_1, e_2, \dots, e_n\}$.

Extending the above defined d by linearity and the graded Leibniz rule, one gets a differential:

$$d : \bigwedge^i \mathfrak{g}^* \rightarrow \bigwedge^{i+1} \mathfrak{g}^*$$

satisfying $d^2 = 0$. The cohomology associated to the cochain complex $(\bigwedge^* \mathfrak{g}^*, d)$ is known as the **Lie algebra cohomology** of $\mathfrak{g}$. In many cases, the Lie algebra cohomology of $\mathfrak{g}$ coincides with the de Rham cohomology of quotients of G by its cocompact discrete subgroups, including the case when G is compact or nilpotent.

Notation for Lie Algebras

For the convenience of notation, we use a symbol like $(0, 0, 0, e^{15}, 0, e^{13})$ to denote the 6-dimensional Lie algebra $\mathfrak{g}$ defined by a basis $\{e^1, \dots, e^6\}$ of $\mathfrak{g}^*$ satisfying:

$$de^1 = de^2 = de^3 = de^5 = 0, \quad de^4 = e^1 \wedge e^5, \quad de^6 = e^1 \wedge e^3$$

Symplectic Lie Algebras

We say (G, ω) is a **symplectic Lie group** or $(\mathfrak{g}, \omega)$ is a **symplectic Lie algebra** if $\omega \in \bigwedge^2 \mathfrak{g}^*$ is a d -closed and non-degenerate 2-form on $\mathfrak{g}$. For a symplectic Lie group or a symplectic Lie algebra, its dimension must be an even number $2m$.

Primitive Forms

For any $0 \leq k \leq m$, we can define the space of **primitive k -forms** as:

$$\mathcal{P}^k \mathfrak{g}^* = \left\{ \alpha \in \bigwedge^k \mathfrak{g}^* : \omega^{m-k+1} \wedge \alpha = 0 \right\}$$

We have the well-known **Lefschetz decomposition**:

$$\bigwedge^k \mathfrak{g}^* = \mathcal{P}^k \mathfrak{g}^* \oplus (\omega \wedge \mathcal{P}^{k-2} \mathfrak{g}^*) \oplus (\omega^2 \wedge \mathcal{P}^{k-4} \mathfrak{g}^*) \oplus \dots$$

and the **Lefschetz isomorphism**:

$$\omega^{m-k} : \bigwedge^k \mathfrak{g}^* \cong \bigwedge^{2m-k} \mathfrak{g}^*$$

Decomposition of the Exterior Derivative

One can easily show that:

$$d : \mathcal{P}^k \mathfrak{g}^* \rightarrow \mathcal{P}^{k+1} \mathfrak{g}^* \oplus (\omega \wedge \mathcal{P}^{k-1} \mathfrak{g}^*)$$

Therefore one decomposes $d = \partial_+ + \omega \wedge \partial_-$, where:

$$\partial_{\pm} : \mathcal{P}^k \mathfrak{g}^* \rightarrow \mathcal{P}^{k\pm 1} \mathfrak{g}^*$$

are first order differential operators mapping primitive forms to primitive forms.

Symplectic Cohomology

The **symplectic cohomology** SH^* of $(\mathfrak{g}, \omega)$ is defined to be the cohomology of the cochain complex:

$$0 \rightarrow \mathcal{P}^0 \mathfrak{g}^* \xrightarrow{\partial_+} \mathcal{P}^1 \mathfrak{g}^* \xrightarrow{\partial_+} \dots \xrightarrow{\partial_+} \mathcal{P}^m \mathfrak{g}^* \xrightarrow{\partial_+ \partial_-} \mathcal{P}^m \mathfrak{g}^* \xrightarrow{\partial_-} \mathcal{P}^{m-1} \mathfrak{g}^* \xrightarrow{\partial_-} \mathcal{P}^{m-1} \mathfrak{g}^* \xrightarrow{\partial_-} \mathcal{P}^0 \mathfrak{g}^* \rightarrow 0$$

Three Types of Cohomology

In this paper, we are concerned with three kinds of cohomologies on $\mathfrak{g}$:

1. De Rham/Lie Algebra Cohomology

$$H^m(\mathfrak{g}; \mathbb{R}) := \frac{\ker \left(d : \bigwedge^m \mathfrak{g}^* \rightarrow \bigwedge^{m+1} \mathfrak{g}^* \right)}{\text{im} \left(d : \bigwedge^{m-1} \mathfrak{g}^* \rightarrow \bigwedge^m \mathfrak{g}^* \right)}$$

2. Primitive Part of De Rham/Lie Algebra Cohomology

$$\text{PH}^m(\mathfrak{g}; \mathbb{R}) := \frac{\mathcal{P}^m \mathfrak{g}^* \cap \ker \left(d : \bigwedge^m \mathfrak{g}^* \rightarrow \bigwedge^{m+1} \mathfrak{g}^* \right)}{\mathcal{P}^m \mathfrak{g}^* \cap \text{im} \left(d : \bigwedge^{m-1} \mathfrak{g}^* \rightarrow \bigwedge^m \mathfrak{g}^* \right)}$$

3. Symplectic Cohomologies

$$\text{SH}_+^m(\mathfrak{g}; \mathbb{R}) := \frac{\ker(\partial_+ \partial_- : \mathcal{P}^m \mathfrak{g}^* \rightarrow \mathcal{P}^m \mathfrak{g}^*)}{\text{im}(\partial_+ : \mathcal{P}^{m-1} \mathfrak{g}^* \rightarrow \mathcal{P}^m \mathfrak{g}^*)}$$

$$\text{SH}_-^m(\mathfrak{g}; \mathbb{R}) := \frac{\ker(\partial_- : \mathcal{P}^m \mathfrak{g}^* \rightarrow \mathcal{P}^{m-1} \mathfrak{g}^*)}{\text{im}(\partial_+ \partial_- : \mathcal{P}^m \mathfrak{g}^* \rightarrow \mathcal{P}^m \mathfrak{g}^*)}$$

Poincaré Duality

The **Poincaré duality** states that the natural pairing:

$$\wedge : \text{SH}_+^m(\mathfrak{g}; \mathbb{R}) \otimes \text{SH}_-^m(\mathfrak{g}; \mathbb{R}) \rightarrow \bigwedge^{2m} \mathfrak{g}^*$$

is nondegenerate if $\mathfrak{g}$ is unimodular.

Maps Between Cohomologies

Since $\partial_+ \partial_- = d\Lambda_\omega d$ and $\ker \partial_- = \ker d \cap \mathcal{P}^m \mathfrak{g}^*$, we naturally have the surjective map:

$$\text{SH}_-^m(\mathfrak{g}; \mathbb{R}) \rightarrow \text{PH}^m(\mathfrak{g}; \mathbb{R})$$

and the injective map:

$$\text{PH}^m(\mathfrak{g}; \mathbb{R}) \hookrightarrow \text{H}^m(\mathfrak{g}; \mathbb{R})$$

On the other hand, as $\text{im} d \cap \mathcal{P}^m \mathfrak{g}^* \subset \text{im} \partial_+$, we also have a natural map:

$$\text{SH}_+^m(\mathfrak{g}; \mathbb{R}) \rightarrow \text{PH}^m(\mathfrak{g}; \mathbb{R})$$

Some Explicit Calculations

The K, F, Q Operators

Let V be a 6 dimensional real vector space equipped with a symplectic form ω . For any 3-form $\phi \in \bigwedge^3 V^*$, we can define $\mathcal{K}(\phi) \in \text{End } V \otimes \bigwedge^6 V^*$ by:

$$\mathcal{K}(\phi)(v) = -\iota_v \phi \wedge \phi \in \bigwedge^5 V^* \cong V \otimes \bigwedge^6 V^*$$

for any $v \in V$. The isomorphism $\bigwedge^5 V^* \cong V \otimes \bigwedge^6 V^*$ is canonical.

If we choose a basis $\{e_1, \dots, e_6\}$ of V with dual basis $\{e^1, \dots, e^6\}$ of V^* , we have:

$$\mathcal{K}(\phi)(v) = - \sum_{i=1}^6 e_i \otimes e^i \wedge \iota_v \phi \wedge \phi \in V \otimes \bigwedge^6 V^*$$

In addition, we can define $\mathcal{F}(\phi) \in \bigwedge^3 V^* \otimes \bigwedge^6 V^*$ by:

$$\mathcal{F}(\phi)(v_1, v_2, v_3) = -2\phi(\mathcal{K}(\phi)(v_1), v_2, v_3)$$

and $\mathcal{Q}(\phi) \in (\bigwedge^6 V^*)^{\otimes 2}$ by:

$$\mathcal{Q}(\phi) = -\phi \wedge \mathcal{F}(\phi)$$

Algebraic Identities

It is clear from their definitions that $\mathcal{K}$, $\mathcal{F}$, and $\mathcal{Q}$ are quadratic, cubic, and quartic in ϕ respectively. Moreover, $\mathcal{K}$, $\mathcal{F}$, and $\mathcal{Q}$ satisfy the following identities:

$$\mathcal{K}(\phi) \circ \mathcal{K}(\phi) = \frac{\text{id}_V}{4} \cdot \mathcal{Q}(\phi) \in \text{End} V \otimes (\bigwedge^6 V^*)^{\otimes 2}$$

$$\mathcal{K}(\mathcal{F}(\phi)) = -\mathcal{K}(\phi) \cdot \mathcal{Q}(\phi) \in \text{End} V \otimes (\bigwedge^6 V^*)^{\otimes 3}$$

$$\mathcal{F}(\mathcal{F}(\phi)) = -\phi \cdot \mathcal{Q}^2(\phi) \in \bigwedge^3 V^* \otimes (\bigwedge^6 V^*)^{\otimes 4}$$

Symplectic Structure and Volume Forms

What we presented so far is independent of the symplectic structure. The symplectic form ω determines a canonical volume form $\omega^3/3! \in \bigwedge^6 V^*$, with which we can view $\mathcal{K}(\phi)$, $\mathcal{F}(\phi)$ and $\mathcal{Q}(\phi)$ as elements in $\text{End} V$, $\bigwedge^3 V^*$ and $\mathbb{R}$ respectively.

Complex Structures from 3-Forms

A notable example is that for ϕ such that $\mathcal{Q}(\phi) < 0$, we can define a complex structure $\mathcal{J}(\phi)$ on V by:

$$\mathcal{J}(\phi) = \frac{2\mathcal{K}(\phi)}{\sqrt{-\mathcal{Q}(\phi)}}$$

The 3-form ϕ is the real part of a complex $(3, 0)$ -form $\phi + i\hat{\phi}$ with respect to $\mathcal{J}(\phi)$. The imaginary part $\hat{\phi}$ is purely determined from ϕ by:

$$\hat{\phi} = \mathcal{J}(\phi)^* \phi = \frac{\mathcal{F}(\phi)}{\sqrt{-\mathcal{Q}(\phi)}}$$

If we define $|\phi|^2$ by:

$$\phi \wedge \mathcal{F}(\phi) = |\phi|^4 \frac{\omega^3}{3!}$$

then we have:

$$\mathcal{K}(\phi) = \frac{|\phi|^2}{2} \mathcal{J}(\phi), \quad \mathcal{F}(\phi) = |\phi|^2 \hat{\phi}, \quad \mathcal{Q}(\phi) = -|\phi|^4$$

Primitive Forms with Symplectic Structure

Under the presence of the symplectic form ω , we usually only consider ϕ that is primitive with respect to ω . In this case, $\mathcal{F}(\phi)$ is also primitive. If in addition $\mathcal{Q}(\phi) < 0$, we know that ω is a $(1, 1)$ -form with respect to the complex structure $\mathcal{J}(\phi)$ and that $|\phi|^2$ is indeed the norm square of ϕ with respect to the metric (not necessarily positive definite):

$$g(\cdot, \cdot) = \omega(\cdot, \mathcal{J}(\phi) \cdot)$$

Basis Representation of 3-Forms

Given a 6-dimensional symplectic vector space (V, ω) , we can choose a basis $\{e_1, \dots, e_6\}$ of V and its dual basis $\{e^1, \dots, e^6\}$ of V^* such that the symplectic form ω takes the standard form:

$$\omega = e^{12} + e^{34} + e^{56}$$

Under such a choice, a general 3-form $\phi \in \bigwedge^3 V^*$ takes the form:

$$\begin{aligned} \phi = & Ae^{135} + Be^{136} + Ce^{145} + De^{146} + Ee^{235} + Fe^{236} + Ge^{245} + He^{246} \\ & + (Ie^1 + Je^2)(e^{34} - e^{56}) + (Ke^3 + Le^4)(e^{12} - e^{56}) + (Me^5 + Ne^6)(e^{12} - e^{34}) \\ & + (Oe^1 + Pe^2)(e^{34} + e^{56}) + (Qe^3 + Re^4)(e^{12} + e^{56}) + (Se^5 + Te^6)(e^{12} + e^{34}) \end{aligned}$$

where $A, B, \dots, S, T$ are constants.

Primitive 3-Forms

And ϕ is primitive if and only if the last 6 coefficients O, P, Q, R, S, T are all zero, namely ϕ takes the form:

$$\begin{aligned} \phi = & Ae^{135} + Be^{136} + Ce^{145} + De^{146} + Ee^{235} + Fe^{236} + Ge^{245} + He^{246} \\ & + (Ie^1 + Je^2)(e^{34} - e^{56}) + (Ke^3 + Le^4)(e^{12} - e^{56}) + (Me^5 + Ne^6)(e^{12} - e^{34}) \end{aligned}$$

Explicit Computation

For the above choice of basis, we compute the expressions of $\mathcal{K}(\phi)$, $\mathcal{F}(\phi)$, and $\mathcal{Q}(\phi)$. The explicit values for $\mathcal{F}(\phi)$ can be found in the source code file `Polynomials.cpp`. For example, the coefficient A is mapped to its symbolic representation `A_hat`.

Main Research Questions

Question 1: Existence and Uniqueness of F-Harmonic 3-Forms

We want to study the problem of existence and uniqueness of $\mathcal{F}$ -harmonic forms, and the long-time behavior of the Type IIA flow on symplectic Lie groups with left invariant data.

Problem: Characterize the existence and uniqueness of $\mathcal{F}$ -harmonic 3-forms in a fixed cohomology class.

By definition, a 3-form ϕ is $\mathcal{F}$ -harmonic if $d\phi = 0$ and $d\mathcal{F}(\phi) = 0$. As an analogue of the standard Hodge theory, we address this in the setup of a symplectic Lie algebra $(\mathfrak{g}, \omega)$ with $\phi \in \Lambda^3 \mathfrak{g}^*$.

Since $d\phi = 0$, we know that ϕ defines a de Rham cohomology class in $H^3(\mathfrak{g}; \mathbb{R})$ for a general 3-form, and that ϕ defines a cohomology class in $PH^3(\mathfrak{g}; \mathbb{R})$ and $SH_{\pm}^3(\mathfrak{g}; \mathbb{R})$ if ϕ is primitive. We want to know: if the cohomology class of ϕ is fixed in the above sense (de Rham for general ϕ , and one of the three kinds of cohomologies for primitive ϕ), does there exist a unique $\mathcal{F}$ -harmonic representative?

Solution Approach For a given symplectic Lie algebra $(\mathfrak{g}, \omega)$, we proceed along the following steps:

Step 1: Compute $H^3(\mathfrak{g}; \mathbb{R})$, $PH^3(\mathfrak{g}; \mathbb{R})$, and $SH_{\pm}^3(\mathfrak{g}; \mathbb{R})$ explicitly.

Step 2: By assuming ϕ taking the form of the general or primitive 3-form representations, reduce the $\mathcal{F}$ -harmonic equation to a system of polynomial equations.

Step 3: Determine the existence and uniqueness of solutions to the system of algebraic equations within each cohomology class.

Question 2: Long-Time Behavior of the Type IIA Flow

Problem: Investigate the long-time behavior of the Type IIA flow on symplectic Lie groups and their quotients.

We want to investigate the long-time behavior of the Type IIA flow:

$$\frac{\partial \phi}{\partial t} = d\Lambda_{\omega} dF(\phi)$$

including whether the flow has long-time existence or finite-time singularities. In its original setup, the initial data is both closed, primitive, and positive. However, our formulation allows us to extend the initial data to be a general 3-form, not necessarily closed or primitive.

We consider such a flow with initial data under three kinds of generalities: 1. A general 3-form 2. A primitive 3-form 3. A closed primitive 3-form

Key Observation There is little difference between the case of general initial data and that of primitive initial data. The Lefschetz decomposition for 3-forms reads:

$$\bigwedge^3 \mathfrak{g}^* = \mathcal{P}^3 \mathfrak{g}^* \oplus \omega \wedge \mathfrak{g}^*$$

For any 1-form $\alpha \in \mathfrak{g}^*$, we have:

$$d\Lambda_\omega d(\omega \wedge \alpha) = d\Lambda_\omega(\omega \wedge d\alpha) = d(d\alpha + \omega \wedge \Lambda_\omega d\alpha) = 0$$

since $\Lambda_\omega d\alpha$ is a constant. This computation implies that $d\Lambda_\omega d\mathcal{F}(\phi)$ has only primitive components, hence the non-primitive components of ϕ are stationary under the Type IIA flow with left invariant data.

Solution Approach To address Question 2, we proceed as follows:

1. **Reduce the Type IIA flow to an ODE system**
2. **Find stationary points**
3. **Analyze the long-time behavior of the ODE system**

Computational Scope

This project computes F-harmonic 3-forms to address the previous two questions with the classification of 6-dimensional symplectic Lie algebras. The classification can be found in the paper “Symplectic or contact structures on Lie groups” (included as `docs/symplectic-lie-groups.pdf`). There, 23 different symplectic Lie algebras are documented.

The description of these Lie algebras in the standard form $\omega = e^{12} + e^{34} + e^{56}$ can be found in `res/input.txt`. The project computes outputs for all these algebras and displays results in generated files.

The explicit values for the $\mathcal{F}(\phi)$ operator for each basis coefficient are implemented in the `src/Polynomials.cpp` file, where each coefficient (e.g., `A`, `B`, `...`, `T`) is mapped to its corresponding symbolic expression (e.g., `A_hat`, `B_hat`, etc.).